Auditable Long-Term Memory: A Deterministic Retrieval Chain Measured at 479/475 of 500 on LongMemEval-S Under the Official Judge

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Abstract

We report 479/500 and 475/500 across two independent full passes on LongMemEval-S (Claude Opus reader via an unpinned CLI alias; a same-day probe resolved it to `claude-opus-5`, §5.3) under the benchmark's official GPT-4o judge, against a published state of the art of 478/500 (95.60%, Chronos High, PwC, arXiv 2603.16862) — one pass above it, one below: indistinguishable from it within the instrument's measured noise band. A second, grok-4.6 reader configuration measured 476/474 on the identical substrate. The system is a deterministic retrieval chain — dense retrieval, cross-encoder reranking, a coverage-first packet compiler, and mechanically extracted reasoning scaffolds, all deterministic code validated by negative controls — with a replaceable LLM reader confined to the final answering step. At margins of a few questions, measurement is the result: the Opus headline pair's entire spread traced to eight verdict-flip rows; cross-judge agreement is 98%; we observed official-judge verdict flips on byte-identical answers; and a maximum-reasoning-effort reader variant scored *lower* (461/465), with 4 of its 19 per-pass losses being the same answer text judged oppositely. We release every reader output, judge verdict, and control receipt, plus an error dossier that attributes all remaining failures — including a strict gold-answer defect, an improvement stage rejected by our own negative control, and the routes that did not work. A full re-score under the official judge costs about $1.28.

1 Introduction

Long-context memory benchmarks are typically reported as single numbers from single runs, judged by an LLM, with no disclosure of run-to-run or judge variance. At the top of a leaderboard, where entries are separated by a handful of questions out of 500, this practice makes the ranking partially an artifact of measurement noise.

This report makes two claims. **A capability claim**: a deterministic, auditable retrieval chain — in which the language model is confined to a replaceable reader role — measures 479/475 per 500 on LongMemEval-S across two independent full passes — pass 1 above the published state of the art (478), the pair indistinguishable from it within measured noise, with overlapping confidence intervals and per-run variance comparable to the gap. **A methodology claim**: the apparatus that produced the number is the more important contribution: every stage below the reader is

*Correspondence: Centennial Defense Systems. Evidence release: `https://github.com/cjchanh/longmemeval-evidence`. Source-tree evidence commits: 477726726 · 04eb64763 · ad71321ea · ba805e575 · 08e5784bb · ad43cc99a · 9b0711c39.

deterministic code validated by negative controls; the promotion rule (what counts as a reportable number) was frozen before the deciding run; measurement noise from both the reader and the judge is quantified and disclosed rather than absorbed into the headline.

The system under test is the retrieval core of Archivist, a local-first conversational memory product. Nothing in the chain is benchmark-specific except the evaluation harness itself. The release reproduces the measurement — materialized packets, scaffolds, reader outputs, judge verdicts, and control receipts — and attests the chain's determinism by receipt; the retrieval, rerank, packet-compiler, and scaffold-operator sources are held (§8).

1.1 Related work

Agentic memory systems (MemGPT, Packer et al., 2023; production systems such as Mem0 and Zep) focus on memory *stores* and management policies; our contribution is orthogonal — determinism below the reader, with the store fixed. Long-term memory benchmarks include LongMemEval (Wu et al., 2024), used here, and LoCoMo (Maharana et al., 2024); we evaluate one benchmark and claim no cross-benchmark generality (§7). The retrieval stack follows the retrieve-and-rerank tradition (Nogueira & Cho, 2019; BGE-M3 family, Chen et al., 2024); reciprocal rank fusion (Cormack et al., 2009) is among the extension-ordering baselines swept in §3.3. On measurement: a self-consistency step (Wang et al., 2023) was built and later withdrawn when internal review showed its two-pass framing was a derived identity (§4.4); the judge-variance study operationalizes known LLM-as-judge concerns (Zheng et al., 2023); the two-pass reporting rule answers the single-run critique of Dodge et al. (2019).

2 Benchmark and evaluation protocol

LongMemEval-S (Wu et al., 2024) contains 500 questions over long multi-session chat histories: 470 answerable questions across six types (single-session-user, single-session-assistant, single-session-preference, multi-session, temporal-reasoning, knowledge-update) and 30 abstention questions (`_abs`) whose correct behavior is to decline to answer. Each question carries a haystack of ∼40–60 sessions with timestamps; evidence for a question may be spread across several sessions recorded weeks apart.

Judge. We use the benchmark's official protocol: GPT-4o (`gpt-4o-2024-08-06`) with the paper's rubric templates. One header sentence of the single-session-preference template was reconstructed from the benchmark's published rubric text; the templates as used are SHA-256 pinned (`9c9d67fab129...`) and released verbatim, so the reconstruction is auditable rather than asserted. Every scoring run executes 12 live positive and 12 live negative control verdicts against a 0.9 pass threshold (preference rows are excluded from the positive control arm because their rubric is the reconstructed one); every run in this report returned 12/12 and 12/12. The harness records the resolved model snapshot per verdict and fails closed on authentication errors. A full 500-row re-score costs ≈ \$1.28.

Comparison point (verified at source, 2026-08-31). The published raw-score state of the art is **Chronos High: 478/500 (95.60%)** (PwC, arXiv 2603.16862, Mar 2026; generator Claude Opus 4.6, LongMemEval judge protocol, per-category table published — their Table 2 and Appendix A, Table 4). Two widely-quoted higher percentages are not raw-comparable: Mastra OM's "94.87%" is a category-unweighted average — their own per-category counts sum to **468/500 raw** (generator gpt-5-mini) — and OMEGA's "95.4%" is likewise task-averaged with raw 466/500, judged by

GPT-4.1 rather than the canonical judge. One higher claim (481/500, "agentmemory V4") exists in an unreviewed personal repository; we note it without treating it as the published bar. Reader tier varies freely across published entries (GPT-4o through Opus 4.6; our headline reader's alias resolved to Opus 5 on a same-day probe, §5.3); we follow the field convention of reporting our strongest configuration alongside the full reader ladder.

3 System architecture

The chain has five stages. Stages 1–4 are deterministic: byte-identical outputs across runs are enforced by verification gates, not assumed (verified on the pinned environment; hardware and library versions are recorded in the release).

3.1 Dense retrieval. Per-question dense embedding retrieval over the haystack produces a ranked candidate pool (local inference; pool ceiling: 468/470 questions have all gold sessions somewhere in the pool).

3.2 Cross-encoder reranking. A cross-encoder (`BAAI/bge-reranker-v2-m3`) scores each candidate session against the question; session score is the maximum chunk score (chunk = date header + turn). Scoring is rank-deterministic (fp16, fixed batch, tie-break by prior rank).

3.3 Coverage-first packet compilation. For each question a reading packet is compiled under a hard budget (16 sessions / 40k proxy tokens): the dense baseline top-10 is preserved as a byte-identical prefix (invariant R1), remaining slots are filled in cross-encoder order with term-preserving extractive compression, and budget overruns drop the lowest-ranked extension first. The compiler is validated by determinism (two compiles byte-identical), a gold-permutation negative control (shuffled gold answers must not change output), a question-ID sabotage control (renamed IDs must not change output), and budget accounting. Packets are gold-complete for 462/470 questions; a sweep of all five extension-ordering policies confirmed the shipped ordering is globally optimal (462 vs. 454 best alternative).

3.4 Deterministic reasoning scaffolds ("facts"). For questions whose type benefits from mechanical reasoning (240 of 470), the chain emits a deterministic evidence index computed from the packet: session chronology, a canonical user-statement recency domain with clock-granular tie keys, a value-preference invariant (statements carrying candidate values outrank bare recency), candidate enumeration with quantity roll-up for counting questions, and granularity/unit and event-anchor rules (contract `v3.4-facts-20260831`). Scaffolds are code with in-code coupling invariants that raise on internal inconsistency. The scaffold is gold-blind and question-ID-independent by construction.

3.5 Replaceable reader. The packet, scaffold, and question go to an LLM reader. The reader is deliberately replaceable; we measure the chain under three readers (§5.4). The headline pair uses the Claude Opus reader over the CLI lane (§5.1); `grok-4.6-high` and GLM-5.3 are the secondary readers (§5.4).

4 Measurement methodology

4.1 Two-pass rule (pre-registered). A nondeterministic reader means a single full run at a one-point margin is not a reportable number. Our promotion rule, adopted two days before the deciding experiment: a headline requires two full passes agreeing, or one number with a stated $\pm$ spread from measured flip variance.

4.2 Judge variance is measured, not assumed. We re-judged both passes of the grok-4.6-high pair with a second judge on identical rows: agreement 98.2%/97.8%, the second judge strictly harsher. We also observed the official judge returning opposite verdicts on a byte-identical answer across runs. At these margins the judge is the measurement floor; we log such rows rather than resolve them by re-rolling.

4.3 Negative controls at the improvement layer. Every proposed improvement runs against a 60-row control set of baseline-correct questions before integration. This rule has teeth: a verifier stage that recovered 3 previously-wrong rows was *rejected* because the control replay showed it flipped 11 of 59 correct drafts to wrong (§6.2). The frozen acceptance rule — zero control flips — blocked it from ever touching the headline, while the reader-tier upgrade in this report passed that same control with zero flips before its full pair was launched.

4.4 Self-consistency tie-break (built, then withdrawn). The raw pair's spread came from eight verdict-flip rows, and a tie-break rule frozen before any additional read gave each flip row three more reads with a five-vote majority. The resulting number was WITHDRAWN from any claim: internal review showed the construction wrote one derived verdict into both passes (an identity, not two agreeing measurements) and its flip-row selection was judge-conditioned. The artifacts remain released as a record; no self-consistency number is claimed anywhere in this report.

4.5 What we did not do. No per-question prompt tuning against gold; no judge re-rolling; no selective reporting (failed routes are in §6); no counting of gold-defect adjudication in any headline number.

5 Results

5.1 Headline

Configuration	Pass 1	Pass 2
v3.4, Claude Opus reader (CLI lane)	**479/500**	**475/500**
v3.4, grok-4.6 high (raw pair)	476/500	474/500
v3.4, grok-4.6 xhigh (agentic transport)	461/500	465/500
v3.4, Claude Opus reader — ex-dossier (8 dossier rows removed, §6.3)	474/492	471/492
v3.4, grok-4.6 high — ex-dossier (same 8 rows removed)	475/492	474/492

Ex-dossier, grok leads Opus 475/474 to 474/471: the Opus margin on the full set is carried by gold-idiom reading on dossier rows (§5.3, §6.3), not by evidence reading on ordinary rows. Both are reported because the headline is the official-judge score and the ex-dossier pair is what the evidence supports.

The Opus pair is the headline: pass 1 exceeds the published SOTA (478) on the raw score; the pair's both-pass-stable floor is 473 (vs. the grok pair's 471); pass 1 is perfect on three of six answerable types (single-session assistant, user, preference). Its 8 flip rows (6 lost — including 2 abstentions and 1 dossier-listed flip-prone row — 2 gained) match the flip-noise structure of every other pair here. Under the pre-registered two-pass rule this is **indistinguishable from the published SOTA within measured noise, one pass above it**, not a clean beat: we state 477±2 and decline the stronger claim our own bar forbids (a pre-release re-judge of pass 1 under the same official judge returned 478 — three verdict flips on identical text, §8 — so even "one above" is a judge roll). The two rows §6.3 documents individually — the $720 workshop row and the 15-weeks row — sit inside this pair as well: one is scored correct because this reader counts a user-stated $500 fee whose event is dated outside the question's four-month window (a boundary row, not a fabrication — §6.3), one is lost to fidelity (the evidence-faithful answer contradicts defective gold) — net zero per pass. Adjudicating the single strict defect raises the Opus pair to 480/476 (479+1 / 475+1); the pair stays within noise of the published state of the art in every stance. Judge controls 24/24 in both passes; a 25-row probe preceded the pair (11/25 stable-wrong recovered, one row no prior reader had ever solved).

The xhigh variant is a measured regression (−15/−9), reported in full. Its gate evidence had looked favorable — a 25-row probe on the high pair's wrong set recovered 9, and a 60-row negative control on baseline-correct rows showed zero flips — but the full pair exposed the gates' limits: only 4 probe recoveries held, and the true correct-row flip rate (∼3.4%) was small enough that a 60-row control had a ∼13% chance of showing zero flips. Row-level attribution of pass 1's 19 losses: 4 are judge flips on answer text identical to the high run's, and 15 share one over-skepticism signature — wrong abstentions on answerable rows, over-filtered counts, hedged golds. More reasoning effort made the reader second-guess correct evidence. (One v1 loss was a staleness artifact, not a reader loss: the reader resumed one row eleven minutes after the judge had scored it on an empty response; a surgical v2 re-judge under fresh 12/12 controls restored it, 460 → 461.)

Raw grok-pair detail: answerable 450/470 and 447/470; abstention 26/30 and 27/30. Judge controls passed in every scoring run. Under the gold-defect adjudication scenario (§6.3) the grok pair reads 477/475 (476+1 / 474+1, the single strict defect); we report that as a scenario, never as a score.

5.2 Per-question-type breakdown (official GPT-4o judge)

Question type	n	Raw p1	Raw p2	xhigh p1 / p2	Opus p1 / p2
single-session-assistant	56	56 (100.0%)	56 (100.0%)	55 / 55	56 / 56
single-session-user	64	63 (98.4%)	63 (98.4%)	61 / 60	64 / 64
knowledge-update	72	70 (97.2%)	71 (98.6%)	69 / 70	69 / 70
single-session-preference	30	29 (96.7%)	29 (96.7%)	25 / 25	30 / 28
temporal-reasoning	127	122 (96.1%)	122 (96.1%)	121 / 123	123 / 123
multi-session	121	110 (90.9%)	106 (87.6%)	103 / 106	110 / 109
abstention	30	26 (86.7%)	27 (90.0%)	27 / 26	27 / 25
Total	**500**	**476**	**474**	461 / 465	**479 / 475**

Five of six answerable types are pass-stable to within one question; the raw pair's entire answerable spread is one type (multi-session), partially offset by knowledge-update. Against the published SOTA's category table on its own abstention-folded basis (n: KU 78, MS 133, SSA 56, SSP 30, SSU 70, TR 133 — their Appendix A), our pass-1 deficit is −3 knowledge-update (75/78 vs. their 78/78), offset by +2 multi-session (120/133 vs. 118/133), +1 single-session-user (70/70 vs. 69/70) and +1 temporal-reasoning (128/133 vs. 127/133), with assistant and preference level — net +1, which is 479 vs. their 478. Pass 2 carries the same −3 knowledge-update and the same +1 single-session-user and +1 temporal-reasoning, ties on multi-session (118/133), and loses −2 on preference (28/30) — net −3, which is 475 vs. 478. We therefore exceed them on multi-session in pass 1 and match it in pass 2; the pass-to-pass difference is preference, not multi-session.

5.3 Ablation ladder (single ruler: official GPT-4o judge)

Stage	Score /500
Dense retrieval + flat reading (Flash reader, v2 prompts)	452/499[1]
+ chain, v3.3 operators, Sol reader	468
+ cross-encoder rerank (v3, grok reader)	473
+ v3.3 operators (official pair)	471 / 470
+ v3.4 operators (official pair, grok high)	476 / 474
+ reader xhigh + agentic transport	461 / 465 (regression)
+ reader tier: Claude Opus, same substrate	**479 / 475**

Ex-dossier (all eight gold-defect/boundary rows removed from both readers): grok leads 475/474 vs. Opus 474/471 — Opus's net margin over grok is carried by gold-idiom reading on dossier rows, not by evidence reading on ordinary rows. The Opus reader is the CLI alias `opus` (claude-cli lane, subscription), not a pinned snapshot: no run artifact records the run-time resolution, and a same-day probe (2026-09-01, Claude Code CLI 2.1.258; the runs used 2.1.252) resolved `opus` to `claude-opus-5` — one model generation newer than the Chronos comparator's Opus 4.6. The grok and GLM readers are named models. Three of Opus's abstention credits (`2133c1b5_abs`, `c8090214_abs`, `f685340e_abs`) are premise-repair-then-answer responses the official judge accepts and the GLM cross-judge rejects; under strict abstention the pair reads 476/472 — still within noise of the published bar. [1]One row of the legacy baseline failed its reader lane and was never re-read; counted as scored-out-of-499 rather than imputed.

5.4 Reader robustness

On identical packets and scaffolds: **Claude Opus 479/475**; grok-4.6-high 476/474; GPT-5.6 (Sol) 455 — measured decomposition versus its prior-contract 468: −5 per-run reader variance on byte-identical non-scaffold rows, −5 negative scaffold×reader interaction, −3 abstention; judge controls 24/24. DeepSeek V4 Pro scores 436/500 (407/470 answerable) with the best abstention behavior of any reader measured (29/30). A 30B floor probe (Nemotron-3.5-Lightning) collapses to 93/500 on the identical gold-complete packets — the chain does not rescue a reader below a capability threshold. We controlled for a serving artifact: correctness barely varies with packet size (28% vs. 15% across size buckets, vs. 90/86% for V4 Pro), and a question-first re-prompt of a failed row was still ignored in favor of distractor-session content — the collapse is model-side, not provider truncation. A gemini-3.7-flash run reached a 189-row partial (a type-ordered prefix, not a sample: single-session-user 64/64, multi-session 73/83, preference 21/30; no other types) and is reported as a bounded signal only. A GLM-5.3 run on the identical substrate scores 471/500 (445/470 answerable, 26/30 abstention; official GPT-4o judge, controls 12/12) — a single-pass curve datapoint per the two-pass rule, not a headline. A Kimi K3 run (the long-context-specialist family) scores 465/500 (444/470 answerable, 21/30 abstention; same judge, controls 12/12) — long-context specialization does not transfer to memory-benchmark dominance on this substrate. Together the fixed substrate exposes a full reader-capability curve (93 → 436 → 455 → 465 → 471 → 476 → 479): the packet and scaffold layers carry the score for capable readers, and reader choice moves it by tens of points across tiers but only a few points within a tier. The v3.4 scaffold contract is tuned to the grok reader family (§7).

6 Error analysis: every remaining failure, attributed

Of 470 answerable questions, 25 fail in at least one pass of the grok-4.6-high pair (pass 1: 20, pass 2: 23; 18 wrong in both). The buckets below attribute those failures; the per-bucket counts are approximate and do not form an exact partition of the union, since a flip row can change bucket between passes.

6.1 Reader-cognition residue (∼4 rows). The packet contains the correct evidence; the reader commits to an evidence-backed distractor. Two verifier designs failed to fix this honestly (§6.2). The residue is question-boundary semantics (does a stated *plan* update a location? does the offered property count as "viewed"?), not evidence selection.

6.2 The verifier our own control rejected. A cite-or-revise verifier recovered 3/12 winnable wrong rows — and the mandatory negative control showed it revised 19/59 baseline-correct drafts, flipping 11 to wrong while repairing none. Blocked by the pre-committed acceptance rule. A second design (draft-blind candidate enumeration + rule-based adjudication) recovered 0/9: on the residue rows the adjudicator reaches defensible-but-not-gold readings. Both failures are published; they map the limit of prompting-side repair. They motivate an evidence-testimony layer as future work.

6.3 Gold-answer defects (1 strict, 7 boundary). Documented row-by-row in the released dossier. Strict B: gold says "15 weeks" between two user-dated events that are 81 days (11 weeks 4 days) apart; wrong in all four passes of both reader configurations. Boundary — the $720 workshop row (published as strict; reclassified after re-auditing the released packet): gold requires a $500 workshop fee, and the user states it (session `answer_826d51da_2`, 2023/02/26 13:37, "I paid $500 to attend"), so nothing has to be invented to reach gold. What is disputed is window membership: both dated mentions place the workshop on March 15–16, after the 2023/02/26 question date, attended in the past tense. The grok reader excludes the fee and answers $220 → 0 in both passes; the headline Opus reader includes it and answers $720 → correct in both passes; a weaker gemini-3.7-flash probe likewise included it and was scored correct. The judge rewards including the out-of-window item because gold includes it — the disclosed window-inclusion point inside 479/475 (§5.1), not a fabrication point. Six further rows are boundary-semantic (defensible readings on both sides); we claim nothing for them. One earlier strict candidate was *downgraded* to boundary after an independent-reader control showed the intended referent was answerable by premise repair. The Chronos authors independently document a defect on question `6d550036` (their §3.5, alongside a judge-variability example on `75f70248`), consistent with our dossier.

6.4 Retrieval ceiling and coverage, measured precisely. One row's gold never reaches the candidate pool; three rows lost an in-pool gold session to packet budget ordering (no global ordering recovers them without losing more). An entity-linking retrieval arm (alias mining + query expansion) was built, tested, and retired when measurement showed zero of its added sessions were gold — the presumed alias-mismatch losses do not exist in this benchmark. Remaining failures are measured flip noise.

7 Limitations

- **No held-out evaluation.** Every component that moves the score — the v3.2→v3.4 operator ladder, the extension-ordering sweeps, the packet budgets, and the 60-row negative-control set — was developed against the same 500 LongMemEval-S questions the headline is measured on, and the gold-defect dossier records per-row inspection of gold answers during development. Nothing in this report is a test-set number in the held-out sense. At a one-to-four-question margin over the published state of the art this is the dominant threat to validity; the mitigations we do have (deterministic stages with negative controls, two independent full passes, every verdict released) bound variance, not overfitting. A held-out run on a disjoint question set is the right next measurement and has not been done.

- **Margins at this level are not statistical superiority.** Wilson 95% intervals: our 479 [93.7%, 97.2%] and 475 [92.7%, 96.6%] (grok pair: 476 [93.0%, 96.8%], 474 [92.5%, 96.4%]) vs. Chronos's 478 [93.4%, 97.1%] — overlapping almost entirely. (The xhigh pair sits lower: 461 [89.5%, 94.2%], 465 [90.4%, 94.9%].) Nearby entries' intervals overlap substantially; no comparison entry publishes per-question verdicts, so McNemar is not computable. Our number is reproduced with variance disclosed; comparison entries are single runs.
- **The judge is the floor.** ∼2% verdict variance between judges, and observed flips on identical answers, bound what any entry at these margins can honestly claim.
- **Scaffold–reader coupling.** The v3.4 scaffold is tuned to one reader family; the exact number moves a few points across readers, though the chain's ranking of readers is stable.
- **The headline reader's provenance is probed, not recorded.** The Opus reader ran through an unpinned CLI alias; no run artifact records the model it resolved to, and the same-day probe that resolved it to `claude-opus-5` ran on a later CLI build than the passes (§5.3). The grok and GLM readers are named models. Any re-run of the headline pair should pin the reader snapshot in the run receipt.
- **One benchmark.** No cross-benchmark generality claim.
- **A claim in our own released dossier was wrong and is corrected here.** The dossier originally classed the $720 workshop row as gold-unreachable without fabrication, on the claim that no $500 workshop fee appeared in the evidence; a mechanical re-audit of this release's own published packets found that fee stated by the user, so the row is a boundary row (§6.3), the strict-defect count is 1 rather than 2, and the adjudication figures in §5.1 were recomputed. No verdict changed and the headline 479/475 is unaffected — only the classification of a row that was already scored as measured. The error and its correction are recorded in the dossier rather than removed from it.

8 Reproducibility

The Chronos and Mastra comparator sources are pinned in the release; OMEGA is not. The Chronos paper (arXiv 2603.16862, PDF and abstract page) and the Mastra Observational Memory page that supply the 478/500 and 94.87% / 468 raw comparator figures are stored with SHA-256 digests and a fetch timestamp under `eval/dense_chain_v32_20260830/comparators/` (see its `MANIFEST.md`), so the comparison rows in §5.1 can be checked against bytes the release carries. The OMEGA figure quoted in §5.1 has no pinned source yet; that gap is recorded in the same manifest.

Released with this report at `https://github.com/cjchanh/longmemeval-evidence` (MIT). A fresh clone verifies every `RELEASE_MANIFEST.md` row and re-derives every headline count from the released verdicts without an API key. The evidence commits on the title page anchor to the source tree the release was exported from. The release holds all reader outputs, all judge verdicts and control receipts for every run in the ladder, the judge harness (frozen rubric, SHA-pinned templates, fail-closed transport), the tie-break rule and per-vote records, attribution tables, the gold-defect dossier, and run scripts. Re-scoring any run under the official judge costs ≈ $1.28 and requires only an OpenAI API key. The released/held boundary is frozen in `RELEASE_MANIFEST.md` (one SHA-256 per released file, regenerated deterministically by `scripts/build_release_manifest.py`): released are all reader checkpoints, judge verdicts, control receipts, agreement matrices, the dossier, the judge harness with its tests and rubric source, and the reader-lane scripts; held are the retrieval, rerank, packet-compiler and scaffold-operator sources. The materialized packets and the v3.4 scaffolds the headline pair consumed are released under `eval/dense_chain_v32_20260830/packets_materialized/` (gzip; sha256 in its `MANIFEST.md`), and `scripts/materialize_packets.py` re-derives every packet byte-for-byte from the released allocation and the MIT benchmark data, so stage 5 (reader) and the judge are re-runnable by a third party on the exact text we read. The held stages ship as pinned artifacts with verification receipts (determinism, gold-permutation, question-ID-sabotage, budget gates); every judge-side claim is re-derivable from the released checkpoints and harness alone. The benchmark data is MIT-licensed (`xiaowu0162/longmemeval-cleaned`), so the released checkpoints may carry question and gold text. Judge-side reproduction, run before release: the frozen pass-1 reader checkpoint was re-judged the same day in a clean output directory with the identical harness (`gpt-4o-2024-08-06`, resolved snapshot exact, fresh controls 12/12 and 12/12). Result **478/500**: 497 of 500 verdicts agree with the quoted 479; answerable is identical (452/470); the three flips are on byte-identical answer text (`7024f17c` and `031748ae_abs` lost, `a2f3aa27` gained). The official judge's own flip floor (§4.2) is therefore measured within a single judge on the headline pass, and "one pass above" is inside that floor; the claim is the noise band, not the rank. Receipts: `full-v34-opus_pass1/judge_gpt4o_repro_20260901/`. The authoritative verdict directories are `full-v34-opus_pass1/judge_gpt4o_v2/` (479) and `full-v34-opus_pass2/judge_gpt4o/` (475); `full-v34-opus_pass1/judge_gpt4o/` is a superseded partial in which 194 rows were never judged (289/500), retained under a `SUPERSEDED.md` marker rather than deleted.

9 Conclusion

A deterministic, negatively-controlled retrieval chain with the LLM confined to a replaceable reader measures 479/475 per 500 on LongMemEval-S — one pass above the published state of the art, the pair indistinguishable from it inside the instrument's noise band — under a measurement discipline that makes the number defensible at margins where measurement is usually the weakest link. The apparatus generalizes: freeze the promotion rule before the deciding run, control every improvement against what it might break, measure the judge, and publish the failures with the successes.

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- BAAI. *bge-reranker-v2-m3.* (cross-encoder reranker used in the retrieval chain)